The Long Tail, Not the Front Page:
Cold-Start Prediction of Crowd Highlight Salience

Kazuki Nakayashiki Keisuke Watanabe
Glasp Inc.
kazuki@glasp.co kei@glasp.co
Co-first authors (equal contribution).

Abstract

A social highlighter's most useful signal — which passages a crowd of readers marks — exists only for documents people have already read. Can the aggregate crowd salience of a document be predicted from its text before its marks accumulate? Prior work on this data found that off-the-shelf zero-shot language models recover highlight locations *worse* than a trivial lead (position) baseline [2], so we ask whether a model *trained* on the highlight corpus can beat that baseline. Using a pre-registered ladder of models and a by-document cluster bootstrap, we find a small but robust edge: a logistic ranker over sentence embeddings and positional/contextual features beats the lead baseline by +0.044 average precision (95% CI $[+0.029, +0.058]$ from a by-document cluster bootstrap; clears a pre-registered margin δ=0.03 in 97% of resamples, and is stable across repeated runs of the same dense-document pool). Two closely related unsupervised extractive baselines (centroid, LexRank-style centrality) *lose* to lead, and the trained model beats them by +0.108, so the edge is not recovered by these generic unsupervised proxies — it reflects learning from real reader marks. In product terms the gain is substantial: precision@3 rises from 0.25 to 0.39 (+0.14 absolute, +55% relative) and the model has higher per-document AP than lead on 69% of documents. An ablation shows both the raw embedding (+0.014) and training augmentation (+0.010) contribute, each with a positive bootstrap CI. The edge is not a temporal-generalization failure, and we find no evidence that content drift or near-duplicate leakage explains it (removing near-twins *raises* it). A standardized regression shows the advantage is governed mainly by document popularity (lower popularity, larger edge) and by label reliability. It nearly vanishes only on the most popular content; reading both the model's and the baseline's per-cell scores, that is where the *lead baseline strengthens*, not where the model weakens. Because our evaluation conditions on documents that eventually accumulated readers, these results are best read as a retrospective cold-start simulation; within that scope, day-zero highlight prediction is most feasible on low-popularity, long-tail documents and least useful on the most popular, front-page-like content where the lead baseline is already strong.

1 Introduction

A social web highlighter accumulates, for each document, the spans many readers chose to mark. Aggregated across readers, these marks are a crowd-salience map and the substrate for popular-highlight displays, summaries, and discovery. In prior work we showed this aggregate layer is the *strong* signal in highlighting — the within-document individual signal is a whisper (own-versus-other gap +0.017 average precision), while what a crowd collectively marks dominates [1] — and that personalizing the salience layer does not help: an impersonal salience order is not improved by personal re-ranking, and zero-shot language models predict highlight locations *worse* than a trivial lead baseline [2].

Every application of the aggregate signal shares a structural gap: it needs marks on the *target* document, and a newly published document has none. We ask whether this gap can be closed from text: given a document's content, can a model predict where its crowd will highlight? Because zero-shot prompting is known to fail here [2], and because the lead baseline (mark the opening sentences) is a hard bar — the recommender-systems analogue, the popularity baseline, is likewise hard to beat [8, 9] — we place the discouraging evidence in front and ask the falsifiable question: does a *trained* model clear lead by a pre-registered margin?

The findings are: (i) a trained model beats lead by +0.044 AP (by-document cluster bootstrap), and beats two unsupervised extractive baselines that themselves lose to lead; (ii) the gain is product-meaningful (precision@3 0.25 → 0.39); (iii) we find no evidence that a temporal artifact, content drift, or near-duplicate leakage explains it; and (iv) a continuous regression shows it is governed by document popularity and label reliability, vanishing only on the most popular content, where the lead baseline — not the model — changes. We also document, transparently, that two intuitive mechanistic stories we initially believed (a temporal shift, then thin labels as a document property) were each refined by the next control, while the bootstrap-robust facts persisted.

Contributions. (1) To our knowledge the first demonstration, on real reader crowds, that aggregate cold-start highlight salience is text-predictable beyond the lead baseline by a *trained* model (+0.044 AP; precision@3 +0.14), in a regime where zero-shot language models and unsupervised extractive baselines do not beat lead. (2) An ablation attributing the edge to representation and augmentation, and a robustness battery (temporal fixed-slice, drift, near-duplicate, multiplicity). (3) A standardized regression and per-cell decomposition localizing the edge to low-popularity documents and showing the low gap in the most-popular cell is the *baseline* rising, not the model failing. (4) A methodological note: controls that refined two plausible mechanisms while the robust facts held.

2 Related work

Salience and highlights. Predicting human salience from text is studied in extractive summarization and highlight detection, usually against reference summaries or small annotator panels rather than real reader crowds; the behavioral finding that language models model human salience only weakly [7] is the zero-shot ceiling our trained model is measured against, and reader highlights predict comprehension and interest [10]. Nearest are work on what makes text quotable or memorable [11], automatic pull-quote selection [12], and industry-facing popular-highlight features on e-reader platforms; to our knowledge none predicts a reader crowd's aggregate highlight map on cold documents. **This program.** Our prior papers locate *where* individuality lives (selection, not salience) [1], show personalization does not help at the salience layer [2], and ask whether the within-document crowd is internally factional [3]. Those study the *individual* or the *segmented* crowd; here we predict the *aggregate* crowd before its marks exist — a different object and task. A separate strand studies answer-engine visibility on this platform [4]. **Hard baselines and aggregation.** That a simple baseline (popularity, position) is hard to beat, and that sampled or mis-specified comparisons distort conclusions [8, 9], motivates our pre-registered lead bar and bootstrap discipline; the complementary result that aggregating simple predictors can rival a human crowd [6, 5] frames why the aggregate layer is the promising target.

Table 1: Dataset summary for one evaluation run.

Candidate dense documents sampled (≥ 20 highlighters)	705
$\rightarrow$ evaluation documents (after fetch + anchoring gate)	284
Training-only candidates ($\in [12, 20)$) $\rightarrow$ usable	1,045 $\rightarrow$ 389
Sentences (evaluation)	32,549
Median sentences / document	72 (range 8–400)
Median anchored co-readers / document	21 (range 6–60)
Median lifetime highlighters / document	27 (range 20–592)
Positive-class base rate (top 15%)	0.146
Article fetch yield	$\sim$40–60%

3 Data and task

Source and funnel. We use Glasp, a social web highlighter with hundreds of thousands of active users and millions of highlighted URLs. From that corpus we sample *dense* documents (at least 20 distinct lifetime highlighters) so an aggregate target is stable, fetch each (live HTTP + readability; yield $\sim$40–60%, paywalls/JS/non-HTML excluded), re-anchor co-reader marks to sentences (excluding social/video domains and capping co-readers at 60 per document), and keep documents with at least six anchored co-readers. Each evaluation run yields $\sim$284 usable evaluation documents and $\sim$389 training-only documents (Table 1). Because we condition on documents that *eventually* reached 20 highlighters, the evaluation population is a survivorship-filtered slice, not a true day-zero population; we treat "cold start" as a retrospective simulation and return to this in Limitations.

Anchoring and label. Glasp does not store article bodies, so we fetch and split into sentences and re-anchor each co-reader's stored highlight strings to sentence indices by text matching (substring, then token-Jaccard ≥ 0.6), keeping a reader only if at least half of their spans anchor, following our prior work [1]. Each sentence's crowd count is the number of anchored co-readers; the target is the top 15% of sentences by count (base rate ≈ 0.146; ties in crowd count broken by a fixed per-sentence jitter, so the threshold is well defined even at low reader counts). All models and baselines rank the *same* labels.

Metric and inference. We report average precision (AP) per document and the *advantage* of a model over lead, AP(model)$-$AP(lead), with a 3,000-iteration cluster bootstrap resampling whole documents; "frac>0" and "frac>δ" are the fractions of resamples with advantage above 0 and above the pre-registered margin δ=0.03. For products we also report precision@k. *Pre-registered* before running: the model ladder, δ, the pass rule, and the primary AP(model) $-$ AP(lead) comparison; the pass rule is a point estimate $\geq \delta$ *and* every by-document resample positive (frac>0 = 1). The ablation, unsupervised baselines, precision@k, and the regression are planned robustness and interpretability analyses added after the primary result. (Random AP is 0.186, slightly above the 0.146 base rate: on short documents the expected AP of a random ranking exceeds the positive rate, a finite-list effect, not a bug.) The pipeline runs on private user data and is not released.

4 Models

A pre-registered ladder of class-balanced logistic rankers over sentences: **M0**, position + lexical features (normalized position, a lead indicator, log length, TF-IDF); **M1**, a raw `text-embedding-3-large` sentence embedding (requested at 768 dimensions via the API `dimensions` parameter) + position; **M2**, M0 + derived semantic/contextual features (cosine to the document centroid, degree centrality, cosine to the lead, cosine to neighbors, a novelty score, surface cues); **M3**, the combined model (raw embedding + the M2 feature set) trained on the dense pool *augmented* with training-only documents of moderate popularity ($userCount \in [12, 20)$), which enlarges training without changing the evaluation population. A one-hidden-layer MLP on the M2 features tests nonlinearity (it does not help). Baselines: **random**; **lead** (earlier sentences ranked higher), the bar to clear; and two closely related *unsupervised extractive* baselines, **centroid** (cosine to the document embedding centroid) and **centrality** (mean cosine of a sentence to every other sentence, a degree-centrality / LexRank-style score; on L_2-normalized embeddings it nearly coincides with the centroid baseline, hence their near-identical scores). M0 underperforming lead is expected: a trained ranker that must generalize across documents does not beat the raw within-document position order until richer features are added.

5 Results

5.1 An edge over lead, attributed, and product-meaningful

The advantage over lead grows with representation and the headline model M3 beats lead by +0.044 AP (Table 2). It meets the pre-registered pass rule (point estimate $+0.044 \geq \delta$; frac>0 = 1.0); as a stricter check it exceeds δ in 97% of resamples, and the de-duplicated estimate (§5.4) excludes δ at the CI level. (The margin was a pre-registered decision threshold for the *point estimate*, not a CI-exclusion criterion; we report frac>δ as a stricter descriptive check.) Figure 1 reports the staged pre-registration runs (M0–M3); Table 2 is a separate run of the same small pool used for the ablation (the dense-document pool is small and the runs overlap heavily; §5.4), so M2 differs slightly (+0.028 vs +0.024) across runs. The ablation attributes the M2→M3 gain to both knobs: the raw embedding contributes $+0.014\,[+0.010, +0.019]$ and the augmentation $+0.010\,[+0.002, +0.018]$, each with a positive bootstrap CI. (The last-mile contributions are weakly superadditive — adding each knob alone to M2 gives +0.010 and +0.005 — so these values include the interaction and slightly overcount the separate effects.) Crucially, the two unsupervised extractive baselines lose to lead (−0.065), and M3 beats centrality by +0.108: the edge is not recovered by these generic unsupervised proxies — it comes from learning real reader marks (a supervised baseline trained on summaries is untested; see Limitations). In product terms the gain is large: precision@3 rises from 0.254 to 0.394 and precision@5 from 0.259 to 0.346 (Table 3), and M3 has higher per-document AP than lead on 69% of documents (ties counted as non-wins).

5.2 Not a temporal-generalization failure

Testing on the newest slice of documents, the advantage falls to +0.009 (not significant). But this is *test-set composition*, not train/test time order. Fixing a future test slice T and scoring the same T two ways — trained on the *past* only versus a same-era cross-validation within T — gives statistically identical results (contemporaneous +0.012 vs temporal +0.011; difference $\Delta +0.001\,[-0.008, +0.010]$). (The earlier +0.009 and this +0.011 differ only because they use different test sets: the newest cross-validation fold versus the separately defined fixed future slice T.) There is no temporal-generalization failure; the newest slice is simply a harder *population* (§5.3). We initially read the +0.009 as a temporal shift; the fixed-slice control retracts that.

Table 2: Ablation and baselines (the ablation run: 284 documents, a separate run of the same pool as Runs A–C; advantage = AP(·) − AP(lead), 95% by-document cluster-bootstrap CI). Unsupervised extractive baselines lose to lead; both M3 knobs contribute.

Method	Note	Advantage over lead	frac>δ
centroid (unsup.)	cosine to doc centroid	$-0.065\,[-0.092, -0.038]$	0.00
centrality (unsup.)	LexRank-style	$-0.065\,[-0.093, -0.039]$	0.00
M2	context features	$+0.024\,[+0.009, +0.040]$	0.24
M2 + augmentation		$+0.029\,[+0.015, +0.043]$	0.43
M2 + embedding		$+0.034\,[+0.018, +0.049]$	0.68
M3	**+ embedding + augmentation**	$\mathbf{+0.044\,[+0.030, +0.058]}$	**0.97**
embedding effect (M3 − M2+aug)		$+0.014\,[+0.010, +0.019]$	—
augmentation effect (M3 − M2+emb)		$+0.010\,[+0.002, +0.018]$	—

Table 3: Product-facing metrics (top-k precision; same run). The model substantially improves top-k precision (a +55% relative gain at k=3).

Metric	lead	M3	M3 − lead
precision@3	0.254	0.394	$+0.141\,[+0.103, +0.180]$
precision@5	0.259	0.346	$+0.087\,[+0.059, +0.116]$

5.3 Where the edge lives: popularity, and the baseline, not the model

A standardized regression of the per-document advantage on document covariates[1] (Figure 3b) shows two robust drivers: lower document popularity ($\beta = -0.32\,[-0.56, -0.14]$) and thicker crowd labels ($\beta = +0.22\,[+0.07, +0.42]$). Recency, a popularity-by-recency interaction, and document length are *not* significant once controlled — so the heterogeneity is a popularity *main effect*, not the interaction a simpler analysis suggested. This reconciles with the per-cell pattern: the high-popularity-and-recent cell concentrates the *most* popular documents (mean lifetime highlighters 81, versus 55 in the high-popularity-established cell), so the continuous popularity main effect alone accounts for its low advantage — no interaction term is needed.

Reading both the model's and the baseline's scores per cell is decisive (Figure 2). The model's absolute AP is roughly stable across the popularity×recency grid (0.34–0.42); what varies is the *lead* AP, which jumps from ∼0.29 on established documents to 0.39 on the most popular, recent ones. So the advantage nearly vanishes there not because the model weakens but because *position becomes nearly as good*: on the most popular content the crowd concentrates on the opening. The honest statement is "the edge is largest where the lead baseline is weak."

Finally, the label-thickness effect at least partly reflects *measurement reliability*: documents with thin crowd labels (few anchored co-readers, a noisier target) show no significant edge ($+0.019\,[-0.010, +0.045]$), while mid and thick labels show +0.05–0.06. A noisier target attenuates any model−lead gap, so +0.044 could be conservative; we caution, however, that thin-label documents are also the closest proxy to a true zero-reader document, so whether the genuinely-new regime is harder to *predict* or merely harder to *measure* remains an open uncertainty rather than

[1] OLS of AP(M3) − AP(lead) on standardized log popularity, recency (the document's median co-reader highlight timestamp), their product, log label thickness (anchored co-readers), and log length; 95% by-document bootstrap percentile CIs. Popularity and label thickness correlate $r = 0.69$.

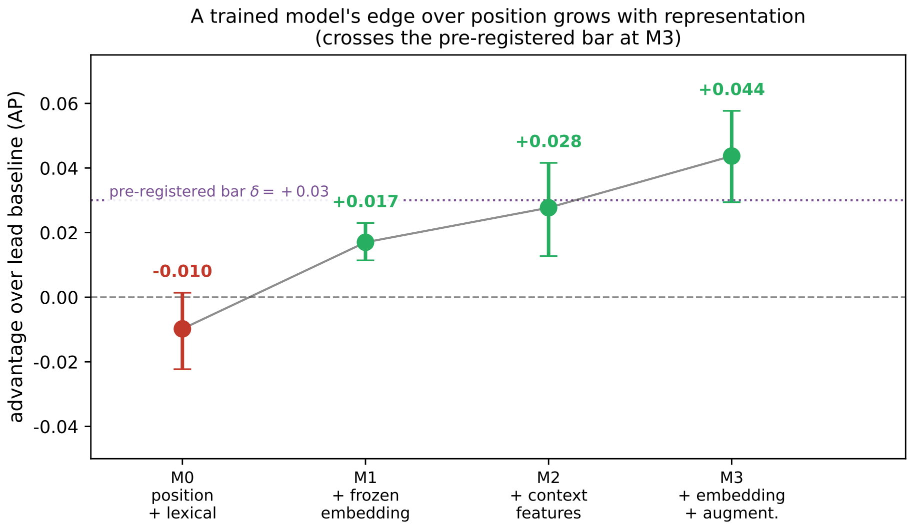

Figure 1: The trained model's edge over lead grows with representation and crosses the pre-registered bar (δ=0.03) at M3 (staged pre-registration runs; Table 2 gives the single-run ablation). Points are mean advantage; bars are 95% by-document CIs.

a settled lower bound.

5.4 Robustness: drift, near-duplicates, anchoring, multiplicity

Content drift. We fetch the current page, whereas readers marked a possibly earlier version. We find no evidence that drift explains the effect: the cleanest (most-recent, least-drifted) documents are not the strongest, and the popularity effect survives controlling recency. We cannot fully exclude drift without archived snapshots, which we leave to future work. **Near-duplicate leakage.** Scoring each test document's maximum embedding-centroid cosine to any training document, only 5% have a $\geq$ 0.95 near-twin, and removing them *raises* the advantage (overall +0.045 $\rightarrow$ clean +0.047 $\rightarrow$ clean-strict +0.053; Table 4); the headline is conservative on a deduped corpus. **Anchoring noise.** Because all methods rank the same labels, anchoring noise cannot by itself create method-specific labels; however, feature-correlated anchoring (e.g. if lead or boilerplate sentences anchor differently) remains a limitation we cannot fully rule out. **Multiplicity.** We computed many advantage contrasts; we therefore lead with the by-document bootstrap CI on the evaluation set (every resample positive) and its stability across pipeline re-runs, rather than the maximum over candidates.

5.5 A methodological note

Two intuitive explanations we first believed were refined by the next control. We read the weak newest-slice result as a temporal-generalization failure; the fixed-slice control (§5.2) showed past-only and same-era training are identical. We then read the weakness as thin/young labels as a document property; the regression and per-cell scores (§5.3) showed label thickness is a *reliability* moderator and that the viral cell reflects a rising baseline. Throughout, the bootstrap-robust facts — the +0.044 edge, its stability across re-runs, the absence of a temporal shift, the unsupervised baselines losing to lead — did not move. We recommend leading with resampling-robust effect sizes and treating each mechanistic "why" as a hypothesis the next control must try to kill.

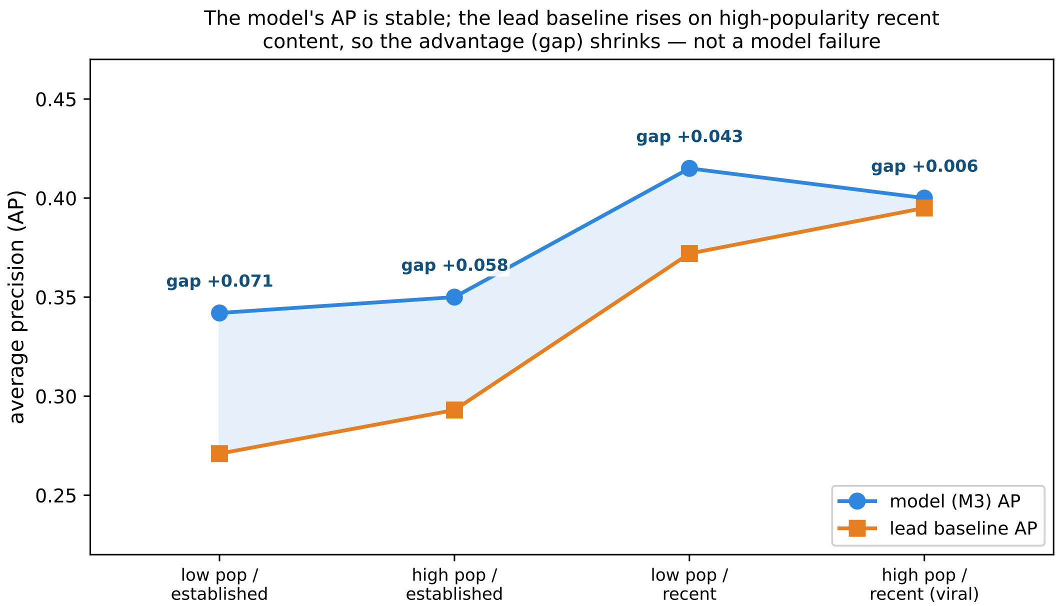

Figure 2: Per-cell model AP and lead AP across popularity × recency (median splits). The model's AP is stable; the *lead* AP rises on the high-popularity, recent cell (which concentrates the most popular documents, mean 81 lifetime highlighters), which is why the advantage (the gap) shrinks there. The edge is largest where lead is weak, not where the model fails.

6 Discussion

Relation to the program. Personal Salience showed the crowd dominates the individual within a document [1]; Selection, Not Salience showed personalization does not help at the salience layer and that aggregating beats personalizing [2]. This paper predicts the aggregate layer those papers identified as strong, on documents lacking marks. There is no contradiction with the +0.017 individual "whisper": that measured a person's history predicting *their own* marks, whereas here the crowd is the *target*. **Product reading.** The cold-start use case — pre-fill popular highlights on a new document — is exactly the low-popularity region, where the edge is positive and bootstrap robust and where organic crowd signal is likely to be slower or sparser. On the most popular content the model and lead are close because position already works and a human crowd arrives fast, so prediction is least needed precisely where it is least distinctive. A day-zero feature is thus most useful on the long tail, with the honest caveat that our evaluation conditions on documents that did eventually reach a crowd.

7 Limitations

Survivorship / simulated cold start. We evaluate on documents that eventually reached ≥ 20 highlighters; this is not a true zero-reader population, and the genuinely-new regime is approximated, not measured. **Label reliability.** Thin labels attenuate the measured edge (§5.3), so +0.044 may be conservative; but the thinnest, most cold-start-like documents are also the noisiest, so whether the genuinely-new regime is harder to *predict* or merely harder to *measure* is unresolved. **Population skew.** Fetchable dense documents skew short and popular; long/niche content and the ~40–60% fetch loss are survivorship filters. **Anchoring and drift.** Live fetching and text

Table 4: Robustness of the headline advantage. The dense-document pool is small (∼288 usable documents); the three runs (A–C) each sample nearly all of it and overlap ∼98%, so they are essentially one evaluation set and their agreement is a consistency check across pipeline re-runs, not independent replication. The primary inference is the by-document bootstrap CI on those documents. De-duplication raises the edge; the temporal control shows no shift.

Check	Run / contrast	Advantage over lead
Stability	Run A (primary)	+0.044 [+0.029, +0.058]
Stability	Run B	+0.042 [+0.028, +0.055]
Stability	Run C	+0.045 [+0.030, +0.059]
Near-dup	Run C, near-twins removed	+0.053 [+0.035, +0.070]
Temporal	contemporaneous − past-only (same slice)	+0.001 [−0.008, +0.010]

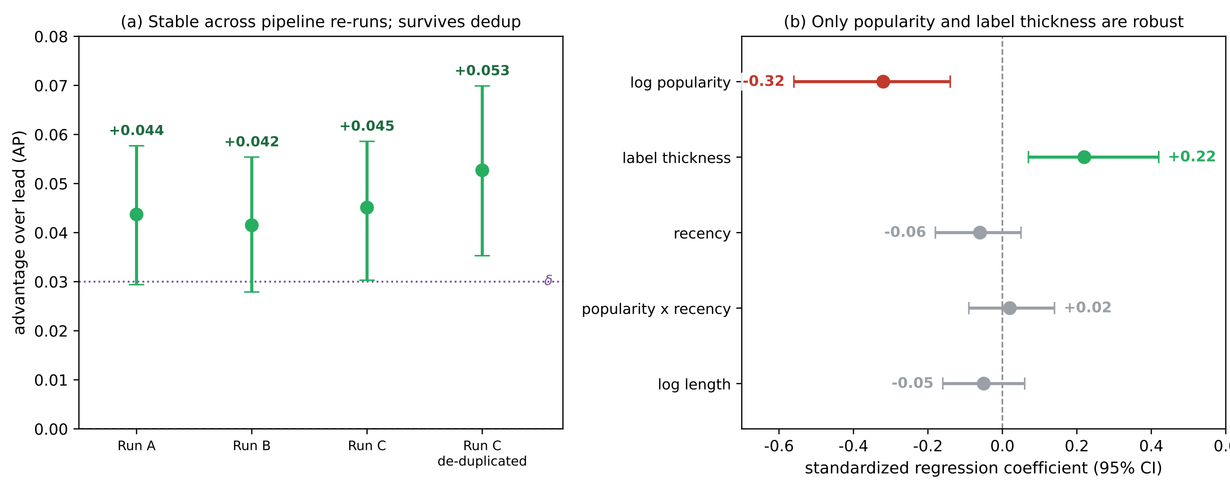

Figure 3: (a) The advantage over lead is the same across three pipeline re-runs of the small document pool (Runs A–C, ∼98% overlapping) and rises after removing near-duplicates; every bootstrap positive. (b) Standardized regression of the per-document advantage: only document popularity (−0.32) and label thickness (+0.22) are robust; recency, the popularity×recency interaction, and length are not.

anchoring introduce noise we control for but cannot eliminate; an archived-snapshot replication and a domain-held-out split are natural next steps. **Label definition.** The top-15% binary target is one choice; a count-weighted target is a natural robustness extension. **Model ceiling.** We test logistic rankers over frozen embeddings; a fine-tuned encoder and a supervised baseline trained on summaries are untested, so M3 is not an upper bound (and the unsupervised result bounds only *unsupervised* alternatives). **Magnitude.** The edge is real and robust but modest; it should be read as "usefully better than position on most documents," supported by the precision@3 gain, not as near-oracle prediction.

8 Ethics

This study analyzes highlighting behavior on a consumer platform under its terms of service. It is internal, aggregate-level analytics rather than experimental intervention: no individual is profiled, no individual-level claims are reported, and no model is trained to predict any individual user — user identifiers are used only to form document-level crowd counts, and documents below a minimum aggregation threshold (fewer than six anchored co-readers) are excluded. We report no individual user's marks, release no per-user or per-document behavioral data, and the extraction/scoring pipeline is not published; highlighting patterns can be identifying, so only bootstrap estimators and aggregate statistics are shared, on reasonable request, and deletion/private-content requests are honored upstream. Predicted highlights are an editorial convenience and carry the usual risk of nudging attention; the popularity gradient we report (weakest on the most popular material) limits the mechanism's reach over the highest-traffic content.

9 Conclusion

On real reader crowds, the aggregate salience of a document is text-predictable beyond the lead baseline by a trained model (+0.044 AP; precision@3 0.25 $\rightarrow$ 0.39), in a regime where zero-shot language models and unsupervised extractive baselines do not beat lead. We find no evidence that a temporal artifact, content drift, or near-duplicate leakage explains the edge; it is governed by document popularity and label reliability, and it shrinks on the most popular content because the lead baseline strengthens there, not because the model fails. Within the scope of a retrospective cold-start simulation, the result places day-zero highlight prediction where it is most useful — the long tail — and the journey reframes a recurring hazard: resampling-robust effect sizes outlive the mechanistic stories attached to them.

Reproducibility

Every effect size carries a 3,000-iteration by-document cluster-bootstrap CI; the model ladder, the margin δ, and the slice definitions were fixed before running. The data-extraction and scoring pipeline runs against Glasp's private user data and is **not** released; per-document results derive from individual highlighting behavior and are not published. The cluster-bootstrap estimator and aggregate statistics are available to researchers on reasonable request.

References

[1] K. Nakayashiki and K. Watanabe. Personal Salience: Highlighting Is Social, but Individuality Lives in Selection. arXiv:2606.09024, 2026.

[2] K. Nakayashiki and K. Watanabe. Selection, Not Salience: The Shape and Limits of Personalization in Social Highlighting. arXiv:2606.10398, 2026.

[3] K. Nakayashiki and K. Watanabe. Factions Within, Uncertain Across: Within-Document Reader Sub-Groups in Social Highlighting. arXiv:2606.11613, 2026.

[4] K. Watanabe and K. Nakayashiki. Disentangling Answer Engine Optimization from Platform Growth. arXiv:2606.04362, 2026.

[5] J. S. Park et al. Generative Agent Simulations of 1,000 People. arXiv:2411.10109, 2024.

[6] P. Schoenegger et al. Wisdom of the Silicon Crowd: LLM Ensemble Prediction Capabilities Rival Human Crowd Accuracy. Science Advances, 2024.

[7] J. Trienes et al. Behavioral Analysis of Information Salience in Large Language Models. Findings of ACL, 2025.

[8] W. Krichene and S. Rendle. On Sampled Metrics for Item Recommendation. KDD, 2020.

[9] Y. Ji, A. Sun, J. Zhang, and C. Li. A Re-visit of the Popularity Baseline in Recommender Systems. SIGIR, 2020.

[10] A. Winchell et al. Highlights as an Early Predictor of Student Comprehension and Interests. Cognitive Science, 2020.

[11] C. Danescu-Niculescu-Mizil, J. Cheng, J. Kleinberg, and L. Lee. You Had Me at Hello: How Phrasing Affects Memorability. ACL, 2012.

[12] T. Bohn and C. X. Ling. Catching Attention with Automatic Pull Quote Selection. COLING, 2020.